

Irish Heat Split: data sources and estimation methodology

Causeway Energies (Causeway Geothermal NI Ltd) · Version 2.0.0 · July 2026

Live site: causewaygt.github.io/irish-heatsplit · Repository: github.com/causewaygt/irish-heatsplit · Correspondence: contact@causewaygt.com

Abstract

The Irish Heat Split estimates the weekly heat balance of the island of Ireland – energy purchased, useful heat delivered, expenditure, indigenous share and emissions – for Northern Ireland, the Republic of Ireland and the island in aggregate, and publishes the estimates daily on a public tracker. Annual fuel-level anchors from national energy statistics [1,3] are distributed across the year by population-weighted heating degree days computed from ERA5 reanalysis [8,9]; the temperature-sensitive component of metered gas demand is estimated by segmented regression following the established gas-to-heat literature [5–7], with within-class (monthly) centring applied to remove seasonal confounding; oil, the island's largest heating fuel, is carried as an anchor-based estimate with a correspondingly wider uncertainty band, accompanied by daily price observation [15,16] which is expressly not treated as a demand feed. Delivered service is obtained from purchased energy through in-situ conversion efficiencies and field-calibrated seasonal performance factors [10]; the indigenous share of delivered heat follows the services-basis convention under which harvested ambient heat is wholly indigenous [24]. Counterfactual (“what-if”) figures recompute the same accounting with one-fifth of delivered heat supplied by ground-source heat pumps. Ground-source capacity accounting combines a bottom-up Northern Ireland register compiled by the authors with the WGC2026 country update for the Republic [12] and European market context [14]. All judgement figures are marked (†) and challengeable; the build pipeline, test suite and every anchor are public.

1 Conventions

- **Sourced versus estimated.** Every figure is either cited to a named public source or marked † as a Causeway Energies estimate. Daggered figures carry a stated basis in the repository and an open invitation to challenge.
- **Purchased energy versus delivered service.** Purchased (input) energy is fuel and electricity bought; delivered service is useful heat after conversion. The distinction is maintained throughout; bars on the tracker labelled “in” and “useful out” correspond to the two bases.
- **Two currencies.** Northern Ireland figures are computed in sterling, Republic figures in euro, and equivalences use the European Central Bank reference rate of the build day [19]; the two price axes of the oil ticker are locked through the same rate so that vertical position is directly comparable.
- **Terminology.** The estimator applied to the gas regression is described as within-class (monthly) centring – the within-transformation of panel econometrics; the tracker's gas-panel label currently references the same computation under an older name, scheduled for alignment with this terminology.

2 Degree days

Daily heating degree days are computed as $HDD = \max(0, 15.5 - \bar{T})$, where $\bar{T}$ is the population-weighted daily-mean two-metre temperature over seven stations (Dublin 0.40,

Belfast 0.20, Cork 0.12, Galway 0.08, Limerick 0.08, Derry 0.06, Waterford 0.06†), from ERA5 reanalysis [8] retrieved through the Open-Meteo archive interface [9] (CC BY 4.0). Three series are maintained – island, Republic-weighted and Northern-Ireland-weighted – so that each jurisdiction's estimates are shaped by its own weather. The 15.5 °C base follows Met Éireann and SEAI practice. An hourly overheating-degree metric for the cooling analysis is scheduled but not yet implemented (§9).

3 The oil anchor – the first hard question

Oil is the island's largest heating fuel – approximately 62% of Northern Ireland households [3,4] and 56.5% of Republic residential heat input [1] – and it has *no demand feed*: deliveries are unmetabled tanker drops into private tanks, unrecorded by any daily statistic. The tracker therefore estimates weekly oil heat from the annual anchor A_e shaped by degree days:

$$W_e = A_e \cdot [(1 - s)/52 + s \cdot H_v/H_v]$$

where $s = 0.72†$ is the weather-driven share of annual demand, H_v the week's degree days and H_v the trailing-year total. The same estimator shapes every unmetered fuel. Its uncertainty is materially wider than the gas estimate's: the anchor itself carries survey and balance uncertainty, the shaping fraction is a judgement, and no daily observation constrains the residual. Figure 1 illustrates the method with an indicative $\pm 20\%†$ band. The Consumer Council daily kerosene survey [16] and the EU Weekly Oil Bulletin with its 2005-onwards price history [15] provide a daily and weekly price pulse alongside the estimate; a price is not a quantity, and the methodology deliberately never treats it as one.

week of year

Figure 1. The oil-anchor estimator: an annual total distributed by weekly heating degree days over a non-weather base, with the indicative $\pm 20\%†$ band. Parameters are illustrative of the island series.

4 Gas-to-heat regression

Metered gas is the island's second heating fuel and its one daily-observed demand. Following the segmented-regression literature [5–7], daily distribution-level demand G is modelled as $G = \alpha + \beta \cdot \text{HDD} + \epsilon$, where β identifies space heating and α the non-weather baseline. Pooled estimation of β is biased by covariates that co-move with winter – holidays, school terms, drifting baseload – so β is estimated after within-class (monthly) centring: demand and degree-day deviations from their own month's means,

$$\hat{\beta} = \sum \tilde{g}_i / \sum \tilde{h}_i^2, \quad \tilde{g}_i = g_i - \bar{g}_m(i), \quad \tilde{h}_i = h_i - \bar{h}_m(i)$$

On the current 204-day record the centred slope is 2.25 GWh per degree day against a pooled 4.03 – the pooled estimate carries nearly half its weight in seasonal confounding rather than weather response. Both estimates ship in the public data file; the centred figure is displayed. Within-month variance explained is 0.66; this is variance explained, not demand explained, and the residual standard error – the true daily uncertainty – is scheduled for display alongside it. Figure 2 demonstrates the estimator on a synthetic record with an injected late-December demand suppression; the identical test ships in the repository and gates every release.

HDD / day

HDD, centred within month

Figure 2. Left: a synthetic daily record with true slope 3.0 and a 30% holiday suppression (amber); pooled OLS drifts while the centred estimator holds the truth. Right: the same data after within-class centring. The test ships in the repository.

4.1 Feed provenance

Gas Networks Ireland data enters by two routes: the public gasconsumption interface, which serves rolling windows of roughly eight days by demand category (daily-metered, non-daily-metered, large daily-metered, power generation) and is walked backwards weekly to assemble a dense daily record; and the quarterly demand CSV published under CC BY 4.0 on data.gov.ie [17], which anchors the regression on a longer stable record. No LDZ/NTS separation of the British kind exists; the non-daily-metered category is used as the distribution-heat proxy. Whether unprefix categories include Northern Ireland exit points is under verification; pending it, the regression is calibrated on the Republic-labelled series and the limitation is stated (§13).

5 Two jurisdictions, one island - the second hard question

Operationally the island is one system twice over – a single wholesale electricity market [18] and one gas network fed through Moffat – but statistically it is two regimes: SEAI/CSO for the Republic [1,20], DfE/NISRA for Northern Ireland [3,4]; CRU-supervised retail pricing in one jurisdiction, Consumer Council observation without a price cap in the other; two currencies. The methodology therefore computes every estimate per jurisdiction from that jurisdiction's anchors, tariffs and degree-day series, and defines the all-island figure as the sum, never the reverse; the property that the three views reconcile is enforced by automated test. Retail tariff anchors are the July 2026 standard-tariff pass (Northern Ireland electricity 32.5 p/kWh, gas 7.5 p/kWh; Republic 36 c/kWh and 11.5 c/kWh); time-of-use tariffs materially improve the heat-pump economics and are noted, not assumed.

6 Delivered service and seasonal performance

Purchased fuel converts to delivered heat at in-situ efficiencies of 82%† (oil boilers) and 85%† (gas). Electric heat-pump routes use seasonal performance factors. The air-source factor is not taken from laboratory ratings: the demand-weighted outdoor source temperature falls out of the degree-day series itself – for heating days $T = 15.5 - \text{HDD}$, hence

$$T_s = 15.5 - \Sigma h_i^2 / \Sigma h_i$$

– and feeds a Carnot-fraction coefficient of performance (fraction 0.38†) at 45 °C† flow with a 0.90† defrost derate, blended on an energy-weighted harmonic basis with a 20%† hot-water share at 55 °C. The result, currently ≈ 2.74 for the Republic and ≈ 2.71 for Northern Ireland, is calibrated to the field-trial medians of the GB Electrification of Heat programme [10] and moves with the weather year. Ground-source systems are carried at SPF 4.0†, the design basis for engineered installations on a stable source [11]; the model makes the mechanism explicit – a stable source temperature is precisely what exempts ground-source performance from the climate ceiling that caps the air-source figure (Figure 3). The break-even SPF against untaxed Northern Ireland kerosene has ranged 3.1–3.6 through July 2026: across the observed price range, modelled air-source performance never clears it, and only ground-source-class performance is in contention.

demand-weighted outdoor temperature, °C ($= 15.5 - \Sigma h^2 / \Sigma h$)

Figure 3. The climate-modelled air-source SPF as a function of demand-weighted source temperature, the 2025/26 jurisdiction operating points, the July 2026 Northern Ireland break-even band, and the geothermal design basis above it.

7 Indigenous share

Each delivered unit of heat inherits the origin of its input on a services basis: imported oil delivers imported heat; grid electricity delivers heat indigenous in proportion to the

jurisdiction's renewable share (Republic 41.3% [1]; Northern Ireland 46%†); and heat harvested from the ground or air by a heat pump is wholly indigenous, following national-balance convention for ambient energy [24]. This accounting is what moves the counterfactual: transferring one-fifth of delivered heat to ground-source raises the island's indigenous share of heat from approximately 14% to 28% in a typical summer week, because three-quarters of every transferred unit becomes locally harvested ambient heat.

8 Electricity

The single electricity market simplifies the island relative to Britain: one day-ahead price feed [18] serves both jurisdictions in both currencies. System-demand and carbon-intensity series from EirGrid are presently unavailable following the July 2026 redesign of the Smart Grid Dashboard; the outage is declared on the tracker rather than bridged, and the replacement endpoint is under investigation.

9 Cooling, waste heat and storage

The island's cooling analysis is data-centre-led. Sectoral electricity is anchored to the CSO metered-consumption series (22% of Republic demand, 29% projected by 2028 [20]) on ≈ 31 TWh†, with 97%† rejected as low-grade heat. Rejection is treated as flat across the year while heat demand follows the degree-day shape; with annual totals normalised, the stranded share is

$$S = \sum_i \max(0, 1/n - h_i/H)$$

– the fraction of a flat supply produced while demand runs beneath it – recomputed from the live record on every build (currently $\approx 34\%$). This is the quantitative case for underground thermal energy storage, whose Irish mirror is heating-dominated: more winter withdrawal than summer banking. Policy attribution is kept precise: the Heat Bill advanced by Cabinet in July 2026 establishes the district-heating framework; the obligation on facilities above 1 MW to assess and implement waste-heat recovery arises under Article 26 of the recast Energy Efficiency Directive [21]. A comfort-deficit (overheating) analysis is deferred: Irish air-conditioning penetration is low, and the British overheating anchors are not imported in its place.

10 Ground-source capacity accounting

No single sourced total exists for island ground-source capacity, so the accounting is assembled from three named layers. Northern Ireland: a bottom-up register of every documented system above 60 kW – named, dated, statused, shipped in full within the public data file – totalling ≈ 0.6 MWth† confirmed, with 500–700† domestic units triangulated from MCS certification counts and a pre-certification-era estimate, and sub-threshold systems logged as documented exclusions. Republic: the WGC2026 country update [12] – 225 MWth, 20,128 systems, 293 GWh/yr, zero deep plants. Europe: comparator capacities from the country-update lineage cross-checked against the EGE 2025 envelope [14], whose 2025 sales table (Ireland 1,409 units; Sweden 26,785) shows the island's gap widening in flow as well as stock; EGE counts units sold where WGC counts capacity commissioned – different measures, both carried, not reconciled. The requirement side converts the counterfactual to hardware at 2,000† equivalent full-load hours: 20% of delivered buildings heat implies $\approx 2,530$ MWth (Republic), $\approx 1,090$ MWth (Northern Ireland), $\approx 3,620$ MWth all-island, against 231.6 MWth† installed. Deep potential in the Permo-Triassic Sherwood Sandstone beneath Northern Ireland is established in Todd et al. [13]; the two regulatory regimes – the Northern Ireland licensing consultation with a register expected ~ 2027 , and the Republic's framework under the 2023 policy statement [22] – are kept distinct.

11 Whole-economy anchors

The closing panel places heat beside transport and power across annual services, expenditure, imported energy and emissions. Republic final consumption (≈ 143 TWh) and emissions are sourced [1]; Northern Ireland totals and all allocations are Causeway derivations kept deliberately round and daggered; the import split allocates the 81.2%-imported share of 220.7 TWh primary energy from the Causeway island Sankey model (2024). The panel's ordering claims – heat rivals transport as the largest service, carries the smallest expenditure per delivered unit, and is therefore the service still dominated by fossil fuel – are enforced by automated test against the anchor set.

12 Validation and uncertainty

- Thirty-seven synthetic tests gate every release: parser fixtures captured verbatim from live run logs; hand-calculated checks of every derivation; the injected-confound regression test of Figure 2; reconciliation of jurisdiction views to the island sum; and the internal logic of the whole-economy anchors.
- Feeds are developed diagnostics-first: on first contact with an unknown format the pipeline logs the raw structure and continues, and parsers are written against that evidence – a discipline that identified the price-history workbook's wide country-column layout before a wrong parser shipped.
- A calibration disclosure – the regression-implied annual gas space heat against the SEAI anchor, with a $\pm 10\%$ acceptance gate – and the display of residual standard error are scheduled improvements; variance-explained statistics are already labelled as such.
- History merges are snapshot-authoritative: current publications win conflicts with archived series, and series breaks are detected and logged.

13 Principal limitations

- The majority fuel is modelled, not metered: oil estimates carry a materially wider band than gas, and no daily observation constrains them.
- The gas regression identifies within-month weather response; genuine between-month structural signal is absorbed by the centring, so truth plausibly lies between the centred and pooled slopes.
- Electricity demand and carbon intensity are pending a replacement EirGrid endpoint; the Great Britain oil comparison is pending a replacement source.
- The jurisdiction of unprefixed Gas Networks Ireland demand categories is unverified; the Northern Ireland domestic ground-source count awaits an MCS export check.
- Whole-economy expenditure and import allocations are the softest anchor layer and are scheduled for hardening on the autumn statistical cycle.
- Summer weeks carry near-zero degree days; small hero figures in June–August are the season, not a fault – the preceding winter's peak week required roughly eight times the energy of a July week.

References

- [1] Sustainable Energy Authority of Ireland, 2025. Energy in Ireland 2025. SEAI, Dublin.
- [2] Sustainable Energy Authority of Ireland, 2022. National Heat Study. SEAI, Dublin.
- [3] Department for the Economy, 2025. Energy in Northern Ireland. DfE, Belfast.
- [4] Northern Ireland Statistics and Research Agency. Continuous Household Survey. NISRA, Belfast.
- [5] Watson, S.D., Lomas, K.J., Buswell, R.A., 2019. Decarbonising domestic heating: what is the peak GB demand? Energy Policy 126, 533–544.

- [6]** Watson, S.D., Lomas, K.J., Buswell, R.A., 2021. How will heat pumps alter national half-hourly heat demands? Empirical modelling based on GB gas demand. *Energy and Buildings* 238, 110777.
- [7]** Sansom, R., 2014. Decarbonising low grade heat for low carbon future. PhD thesis, Imperial College London.
- [8]** Hersbach, H., et al., 2020. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society* 146, 1999–2049. doi:10.1002/qj.3803.
- [9]** Open-Meteo, historical weather interface to ERA5 (CC BY 4.0). open-meteo.com.
- [10]** Energy Systems Catapult, 2023–2024. Electrification of Heat demonstration project: heat-pump performance reporting. ESC, Birmingham.
- [11]** Energy Systems Catapult, 2023. In-situ heat pump performance. ESC, Birmingham.
- [12]** Ireland, J., Blake, S., Pasquali, R., Dunphy, R., Hunter Williams, T., 2026. Country update: Ireland. *Proceedings World Geothermal Congress 2026*, Calgary.
- [13]** Todd, S., et al., 2026. Geoenergy. doi:10.1144/geoenergy2025-057.
- [14]** European Geothermal Energy Council, 2026. EREC Geothermal Market Report 2025 – Key Findings. EREC, Brussels.
- [15]** European Commission. Weekly Oil Bulletin and price-history workbook. energy.ec.europa.eu.
- [16]** Consumer Council for Northern Ireland. Home heating oil price checker (daily survey). consumercouncil.org.uk.
- [17]** Gas Networks Ireland. Gas demand data, via data.gov.ie (CC BY 4.0), and the public gasconsumption interface.
- [18]** SEMOpX. Day-ahead market results. semopx.com.
- [19]** European Central Bank. Euro foreign exchange reference rates. ecb.europa.eu.
- [20]** Central Statistics Office, 2024–2025. Data centres metered electricity consumption. CSO, Cork.
- [21]** Directive (EU) 2023/1791 on energy efficiency (recast), Article 26. *Official Journal of the European Union*.
- [22]** Department of the Environment, Climate and Communications, 2023. Policy statement on geothermal energy for a circular economy. DECC, Dublin.
- [23]** Causeway Energies, 2026. Northern Ireland ground-source register (≥ 60 kW), shipped in the tracker data file. github.com/causewaygt/irish-heatsplit.
- [24]** Eurostat. Energy balance guide: treatment of ambient heat. European Commission, Luxembourg.

Citation forms above are stated to the authors' current records; a verification pass against primary sources precedes any print or archival release, per the project's sourcing standard.